

ALTAY, China

WMO: 510760

Lat: 47.7410N

Lon: 88.0710E

Elev: 737

StdP: 92.78

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.0	-27.1	-33.3	0.2	-29.8	-30.3	0.2	-26.8	5.4	-12.1	4.4	-13.1	1.2	0	0.516

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	31.8	18.3	30.3	17.7	28.8	17.1	19.9	28.7	19.0	27.6	18.2	26.6	2.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.9	13.2	23.7	15.9	12.3	22.8	15.0	11.6	22.1	60.5	28.7	57.3	27.5	54.5	26.7	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.7	6.4	5.4	DB	-32.7	34.9	3.5	1.7	-35.2	36.2	-37.2	37.2	-39.2	38.2	-41.7	39.5	(4)
(5)			WB	-32.8	22.0	3.5	1.6	-35.3	23.2	-37.3	24.1	-39.3	24.9	-41.8	26.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.3	-15.7	-12.2	-3.1	9.0	15.5	20.8	21.9	20.4	14.5	7.2	-3.1	-12.4	(6)	
	DBStd	14.38	5.82	5.76	7.28	5.65	4.67	3.46	2.98	3.45	4.56	5.08	6.84	5.53	(7)	
	HDD10.0	3136	797	622	408	83	9	0	0	0	12	116	394	695	(8)	
	HDD18.3	5078	1055	855	665	282	111	16	4	18	129	346	644	953	(9)	
	CDD10.0	1427	0	0	2	52	181	323	370	322	148	29	1	0	(10)	
	CDD18.3	328	0	0	0	2	24	88	117	81	15	1	0	0	(11)	
	CDH23.3	2938	0	0	0	12	211	783	1031	761	138	2	0	0	(12)	
	CDH26.7	909	0	0	0	1	44	246	348	243	27	0	0	0	(13)	
(14) Wind	WSAvg	2.1	1.3	1.5	2.0	3.1	3.0	2.5	2.1	2.2	2.2	2.2	1.8	1.3	(14)	
(15) Precipitation	PrecAvg	197	15	11	10	13	17	16	25	18	16	16	23	20	(15)	
	PrecMax	339	75	52	53	42	57	57	83	61	55	40	66	64	(16)	
	PrecMin	77	0	0	0	0	0	0	0	1	0	0	2	2	(17)	
	PrecStd	60	14	10	10	10	13	11	17	15	12	9	15	16	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.7	1.5	16.2	25.0	30.3	32.8	34.4	33.5	29.7	23.0	13.8	1.4	(19)	
		MCWB	-2.3	-1.4	7.7	11.7	15.0	18.1	20.9	18.7	15.5	10.0	5.8	-1.4	(20)	
	2%	DB	-3.4	-0.8	12.6	22.0	27.7	31.0	31.9	31.2	26.8	19.5	10.5	-1.4	(21)	
		MCWB	-4.9	-3.3	6.0	10.0	13.9	17.3	19.2	18.3	14.0	8.7	4.1	-3.0	(22)	
	5%	DB	-5.8	-2.7	9.6	19.7	25.7	29.5	30.3	29.3	24.7	17.2	7.9	-3.1	(23)	
		MCWB	-7.1	-4.7	4.5	9.0	13.0	16.8	18.6	17.4	12.9	8.1	3.0	-4.3	(24)	
	10%	DB	-7.6	-4.4	6.9	17.5	23.7	27.9	28.6	27.6	22.6	14.9	5.5	-4.7	(25)	
		MCWB	-8.9	-6.0	2.7	8.2	11.8	16.3	18.0	16.8	12.2	7.2	1.5	-6.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.8	-1.1	8.2	12.6	16.2	19.7	22.4	20.6	16.6	11.2	6.8	-0.8	(27)	
		MCDB	-1.1	1.2	14.5	21.9	27.4	28.7	32.0	30.2	27.6	18.2	12.0	0.7	(28)	
	2%	WB	-4.9	-3.0	6.4	11.1	14.8	18.7	20.6	19.3	15.1	9.8	4.8	-2.8	(29)	
		MCDB	-3.6	-1.0	11.8	19.2	25.2	28.1	29.3	28.4	24.5	17.2	9.7	-1.8	(30)	
	5%	WB	-7.0	-4.6	4.8	9.9	13.9	17.9	19.5	18.3	13.9	8.8	3.3	-4.2	(31)	
		MCDB	-5.8	-2.9	9.3	17.5	23.7	27.2	27.9	27.3	22.8	15.7	7.4	-3.0	(32)	
	10%	WB	-8.9	-6.0	2.9	8.9	12.9	17.1	18.7	17.4	12.9	7.8	1.8	-6.0	(33)	
		MCDB	-7.7	-4.5	6.6	16.1	21.7	25.8	26.7	26.0	21.0	14.1	5.1	-4.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.3	10.0	9.4	10.1	10.5	10.9	11.1	11.8	12.1	9.9	8.0	8.5	(35)	
		MCDBR	9.3	10.7	10.5	12.3	12.2	12.6	13.1	13.6	13.8	12.6	10.2	8.4	(36)	
	5% WB	MCWBR	8.1	8.8	6.5	5.9	5.1	5.3	6.4	6.6	6.1	6.0	6.0	7.0	(37)	
		MCWBR	8.8	10.1	9.8	10.8	11.3	11.4	11.5	12.3	12.8	11.3	9.4	7.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.274	0.303	0.345	0.387	0.375	0.387	0.385	0.364	0.331	0.324	0.301	0.276	(40)	
		taud	2.159	2.112	2.107	2.138	2.249	2.296	2.322	2.351	2.398	2.323	2.213	2.141	(41)	
	Ebn at Noon		787	842	861	861	887	876	872	877	877	816	745	727	(42)	
		Edh at Noon	83	110	131	141	132	127	122	113	98	89	79	75	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.75	2.82	4.25	5.46	6.56	6.88	6.64	5.94	4.69	2.84	1.63	1.31	(44)	
	RadStd		0.11	0.13	0.22	0.33	0.38	0.38	0.36	0.23	0.31	0.27	0.17	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	-0.56	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page